



Implicit and explicit linguistic biases: The influence of social dominance orientation (SDO) upon hierarchical language attitudes

Robert M. McKenzie^{a,*}, Kristofor McCarty^b, Mimi Huang^a

^a *English Language and Linguistics, Department of Humanities, Northumbria University, Lipman Building, Sandyford Road, Newcastle Upon Tyne NE1 8ST, United Kingdom*

^b *Department of Psychology, Northumbria University, Sandyford Road, Newcastle Upon Tyne NE1 8ST, United Kingdom*

Received 13 December 2024; revised 31 January 2025; accepted in revised form 5 March 2025; available online 20 March 2025

Abstract

Research indicates that various social factors can predict language attitudes. However, although social dominance orientation (SDO), the individual's preference for societal hierarchies, has been found to determine evaluations of a range of social groups and socially meaningful topics, its role in predicting more deeply embedded implicit attitudes in conjunction with more deliberative explicit attitudes towards specific language varieties and their speakers is currently unknown. The present research examined the effect of SDO upon English nationals' ($N = 306$) explicit and implicit evaluations of phonological variants indexical of (i) Northern English speech and (ii) Southern English speech in England on competence (Study 1) and warmth (Study 2) dimensions. Regression analysis, controlling for demographic variables, demonstrated that high-SDO predicted negative explicit competence and warmth attitudes towards Northern English speech. Conversely, SDO did not determine self-report Southern English speech evaluations. Likewise, SDO did not predict implicit competence or warmth attitudes. The study findings are discussed in relation to the methodological and theoretical value for (socio)linguists of incorporating SDO measures into research investigating language attitudes at different levels of evaluational awareness.

© 2025 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Keywords: Language attitudes; Social dominance orientation (SDO); Implicit vs explicit language attitudes; Implicit association test (IAT); Individual differences

1. LANGUAGE ATTITUDES

The combinations and frequencies of the words, sounds and grammatical forms a speaker employs carry considerable social meaning and often evoke strong public evaluations regarding the perceived correctness and

* Corresponding author.

E-mail addresses: robert.mckenzie@northumbria.ac.uk (R.M. McKenzie), kristofor.mccarty3@northumbria.ac.uk (K. McCarty), mimi.huang@northumbria.ac.uk (M. Huang).

<https://doi.org/10.1016/j.lingua.2025.103928>

0016-7037/© 2025 The Author(s). Published by Elsevier B.V.

This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

appropriateness of the speech. This is because linguistic variation is socially structured within communities and the variety an individual speaks, composed of specific linguistic features, enables the listener – whether correctly or incorrectly – to index the speaker's membership of a particular dominant or subordinate social, national or ethnic group (Silverstein, 2003). Speaker evaluations are thus indicative of wider social stereotypes and group biases, and knowledge of listener language attitudes can help researchers account for levels of societal prejudice more broadly. Indeed, public attitudes towards different languages/language varieties have important social implications for speakers including, determining job interview outcome, the perceived guilt or innocence of court defendants, schoolteachers' judgments of their students' abilities, the acceptance (or not) of specific immigrant groups and the perceived trustworthiness and persuasiveness of the speech utterance itself (Braber et al., 2023; McKenzie and Gilmore, 2017).

Prior language attitude research has demonstrated that, upon the presentation of speech stimuli, listeners can differentiate between varieties of a single language of which they have some proficiency and are willing to make social judgments about the perceived (community of) speakers under consideration (Garrett, 2010; Levon et al., 2021). The findings from these studies, mainly employing self-report (i.e., explicit) instruments, demonstrated – across many different languages and their respective varieties – that evaluations are often structured according to two primary, non-overlapping and often compensatory dimensions: *competence* (also labelled as status or correctness) and *warmth* (also labelled as social attractiveness or solidarity) (McKenzie et al., 2016). Language attitude research also indicated that (speakers of) linguistic varieties perceived as standard are generally judged positively on competence, and thus rated highly on traits such as wealth and education, whilst (speakers of) forms perceived as non-standard are frequently judged positively on warmth, and rated highly on friendliness, and honesty (Garrett, 2010). The primacy of the competence-warmth evaluative dimensions uncovered by language scientists has also been demonstrated by psychologists investigating individuals' attitudes towards other social groups (Fiske et al., 2007). In short, the compensatory competence-warmth attitudinal distinction appears to be fundamental in human social categorisation (Abele et al., 2021).

In England, research employing self-report language attitude measures has demonstrated that English nationals generally rate the English spoken in the south of England – and in particular Received Pronunciation (RP), and its modern equivalent, Standard Southern British English (SSBE) – more positively on competence than varieties spoken in the north of the country (Coupland and Bishop, 2007; Sharma et al., 2022). Conversely, Northern English speech forms are generally rated more favourably on warmth than Southern English (Hiraga, 2005; Strongman and Woosley, 1967). This evaluative pattern underlines the salience of the binary north–south divide for English nationals and considered reflective of wider, long-standing stereotypes regarding the relative wealth of the south of England, as the centre of power in England and, in turn, the perceived high status of its inhabitants (Taylor, 1991). Conversely, English forms spoken in northern England are typically rated unfavourably for competence due to associations with comparatively high deprivation rates, especially within the post-industrial northern cities (Wales, 2000). In this way, the English spoken in southern England – and especially those speech forms associated with the Home Counties in the wider London area – indexes the higher status, dominant south whilst Northern English speech indexes the lower status, subordinate north of England. Compensatory high warmth Northern English-low warmth Southern English speech evaluations are felt to reflect wider stereotypes of northern English people as humorous and warm and southern English people as more reserved and elitist (Wales, 2006).

Attitude research undertaken within psychology traditionally utilised self-report measures to examine participants' *explicit attitudes*, i.e., deliberative evaluations which the individual can reflect upon and verbalise (Fiske and Taylor, 2021). As a result of fine-grained development of dual-process theories, alongside innovations in computer technology, psychologists also increasingly investigate individuals' *implicit attitudes*, i.e., automatically-activated evaluations (also known as automatic attitudes), which are not open to verbalisation and occur spontaneously upon presentation of the stimuli in question (e.g., De Houwer and Moors, 2007; Gawronski and De Houwer, 2014). Numerous implicit attitudes measures have been developed, the most widely employed of which is the implicit association test (IAT) (Greenwald et al., 1998). Whilst a detailed description is available elsewhere (Nosek et al., 2007), in short, the IAT measures the speed at which participants can pair a given stimuli with evaluative traits. For example, participants who more rapidly associate white faces relative to black faces with positive traits would indicate an implicit bias towards the white faces. With regards to language use, more rapid associations between a phoneme perceived as standard English and positive traits when compared to slower associations between a phoneme perceived as non-standard English and positive traits would indicate an implicit preference for the standard speech form.

Numerous studies investigating implicit evaluations of a range of social category differences – including gender, age, ethnicity and national/regional identity – indicated the IAT is a particularly robust predictor of deeply-embedded prejudice, and especially of implicit biases against, and thus prejudice towards, groups perceived as subordinate and, conversely, implicit biases in favour of groups perceived as dominant (Lane et al., 2007).

The majority of *language* attitude studies also investigated participants' explicit linguistic evaluations. However, more recently, several studies examined participants' implicit language attitudes. The results generally uncovered negative implicit evaluations of non-standard varieties and positive implicit associations with standard forms for speaker competence, including in Belgium (Rosseel et al. 2018), Catalonia (Ianos et al., 2023), Italy (Calamai and Ardolino, 2020), South Africa (Álvarez-Mosquera and Marin-Gutiérrez, 2021), The Netherlands (Dekker et al., 2021), the United States (Campbell-Kibler, 2012) and England (McKenzie and Carrie, 2018). With regards to the latter, McKenzie (ibid.) found, for competence, English nationals were significantly more negative towards (speakers of) Northern English than Southern English speech forms at implicit when compared with explicit evaluational levels. Gender and age differences were also uncovered with younger females expressing less negative self-report attitudes towards Northern English speech.

2. SOCIAL DOMINANCE ORIENTATION (SDO) AND LANGUAGE ATTITUDES

There exists considerable evidence that language-based evaluations are not homogenous within speech communities and researchers have found various individual differences predict linguistic attitudes. In an ever-widening research agenda to identify the listener factors which determine linguistic judgments, language scientists have found listener age (Baker, 1992; McKenzie and Carrie, 2018), gender (Álvarez-Mosquera and Marin-Gutiérrez, 2021; McKenzie, 2008) and national/regional identity (Coupland and Bishop, 2007; Dragojevic et al., 2015), are especially robust determinants of competence and warmth attitudes towards various languages and language varieties at explicit and, recently, implicit levels of evaluation.

Social dominance orientation (SDO) is a further individual difference which psychologists have demonstrated to be a powerful predictor of intergroup attitudes towards numerous dominant and subordinate groups. Drawn from social dominance theory (SDT), which attempts to understand the processes by which group-based social hierarchies are formed and maintained (Pratto et al., 2006), SDO is a status-endorsing ideological variable constituting an individual's level of preference for inequality between social groups within hierarchically-structured societies and, by extension, the degree to which the individual commits to legitimising myths which support status-related (i.e., competence-related) hierarchies and inequalities (Pratto et al., 1994).

SDO has typically been investigated as a unidimensional construct and often operationalised by means of a 14-item scale (the SDO₅) and, more recently, a 16-item scale (the SDO₆) (Ho et al., 2012). Theorists argue that SDO effects upon competence-related attitudes can be divided into three distinct societal group hierarchies; age, gender and arbitrary-set features – in terms of the latter, including social class, region of origin or language usage – which vary substantially in hierarchical severity between and within societies over time (Pratto et al., 2006).

There exists robust evidence from self-report attitude studies that high-SDO individuals generally evaluate (members of) subordinate social groups more negatively but powerful, dominant groups more positively for competence. This effect for high-SDO individuals' explicit competence attitudes has been uncovered in relation to numerous non-language related social groups and socially meaningful topics, including age, gender, ethnic group, race, racial policy, welfare and sexuality (see Kteily et al., 2012; Pratto et al., 2013).

Several studies have assessed the influence of SDO upon both implicit and explicit attitudes. The results generally indicated that SDO predicted explicit but not implicit attitudes, including evaluations of Christians and Muslims (Rowatt et al., 2005), Barack Obama and foreignness (Crawford and Bhatia, 2012), religious persuasion and the gay community (Anderson and Koc, 2015) and gay males (Moor et al., 2019). Other research found contradictory evidence that SDO may, to some extent, influence implicit biases, albeit under very specific conditions. In a laboratory-based study investigating university students' implicit attitudes towards a high-competence ingroup and low-competence outgroup, Pratto and Shih (2000) found high-SDO predicted prejudicial negative implicit biases against the lower competence outgroup only when the context was made salient, i.e., when researchers primed the context for participants. Stern and Axt (2019) found that high-SDO determined both implicit and explicit negative attitudes towards African Americans in the United States. In Australia, Anderson (2018) also uncovered evidence that SDO predicted both negative implicit and explicit evaluations of asylum seekers. Given the somewhat inconsistent findings, researchers have called for further research investigating implicit attitudes towards a greater range of dominant and subordinate groups to help better understand the relationship between SDO and more deeply-embedded biases (Anderson, 2018).

Indeed, considering prior research generally examined SDO effects upon explicit and implicit attitudes towards non-language-based social groups – alongside empirical evidence indicating the greater role which language plays in comparison with visual information in evaluations of social group hierarchies (Kinzler et al., 2009; Rakić et al., 2011) – it is surprising the investigation of SDO as a determinant of attitudes towards *linguistic* stimuli remains scarce, particularly regarding evaluations of standard and non-standard speech forms. Indeed, since all languages and language varieties

are structured systematically, whereby each speech community adheres to its own complex internal linguistic and external social constraints (Labov, 2001), categorisations of speech forms as either 'standard' or 'non-standard' are thus based upon speaker competence judgements rather than any inherent value of their respective linguistic features (Edwards, 2011). Reflective of established social hierarchies, the most powerful and dominant social groups within a particular society typically valorise the speech variety they themselves employ as the sole local/national elite standard form(s) which, in turn, downgrades the speech varieties employed by subordinate groups as non-standard (Milroy and Milroy, 2012). It thus seems possible that social dominance orientation – as a powerful predictor of subordinate and dominant group-based hierarchies – influences competence judgments of linguistic varieties perceived as more or less standard, at least at explicit evaluational levels.

With regards to the previous language attitude research undertaken, in a study involving the self-reported responses of US participants within an imaginary job interview context, Hansen and Dovidio (2016) found that high-SDO predicted greater perceived difficulty to understand a more negatively evaluated Latino English speaker than a more positively rated Asian English speaker and, in turn, uncovered a broader unwillingness amongst participants to hire Latino English speakers. Hansen (2020) also found that high-SDO determined more negative explicit competence attitudes towards non-native users of Polish. In a study involving US nationals, Sumantry and Choma (2021) found that high-SDO predicted more negative self-report competence, but not warmth, attitudes towards several L2 English speech forms when compared with Canadian English.

Taken together, and broadly consistent with findings investigating SDO as a determinant of evaluations of other socially meaningful topics and social groups, the results from the limited language attitude research suggests high-SDO may also determine prejudicial status-related explicit evaluations of speech communities perceived as subordinate. However, there seems a requirement to investigate the effect of SDO upon *implicit* attitudes towards different language varieties. In-depth linguistic research, incorporating both implicit and explicit attitude measures and examining SDO effects together with other potential predictor variables may help clarify the role which SDO plays in listeners' deeply-embedded and deliberative judgments of language varieties and their communities of speakers. Likewise, since SDO is typically assumed to constitute a status-endorsing ideological variable (Pratto et al., 1994), researchers have generally assessed SDO influences upon competence attitudes only. To better understand the influence SDO may have upon listeners' compensatory warmth-competence judgments of different language varieties, it seems worthwhile to conduct research incorporating warmth as well as competence evaluative traits. Given the lack of consistency found regarding SDO effects upon implicit biases towards, and against, a range of dominant and subordinate groups, further information regarding the influence of SDO upon implicit evaluations of linguistic stimuli may help researchers and theorists build up a more detailed picture of the relationship between SDO and social attitudes more broadly.

3. THE PRESENT RESEARCH

The above discussion highlighted the dearth of previous studies examining the role of SDO on language attitudes, including amongst English nationals. The primary objective of the present research was to examine the effect of SDO upon English nationals' explicit and implicit evaluations of Northern English and Southern English speech for competence and warmth. To achieve this, two separate language attitude studies were undertaken: Study 1 explored the effect of SDO on competence evaluations whilst Study 2 explored the effect of SDO on warmth evaluations at both explicit and implicit levels.

As detailed above, there exists considerable empirical evidence that SDO is a robust determinant of individuals' support for competence-related, but not warmth-related, social group inequalities. Moreover, English nationals' self-report competence evaluations of (speakers of) Southern English forms have historically been more positive than (speakers of) Northern English varieties. Recent linguistic research also uncovered implicit biases towards, and against, particular language varieties, including those spoken in England. Accordingly, the following hypotheses were formulated:

Hypothesis 1a. English nationals with high-SDO will hold more positive explicit competence, but not warmth, attitudes towards Southern English speech.

Hypothesis 1b. English nationals with high-SDO will hold more negative explicit competence, but not warmth, attitudes towards Northern English speech.

Hypothesis 2. English nationals with high-SDO will hold more negative implicit competence, but not warmth, biases against Northern English speech when compared to Southern English speech.

4. STUDY 1: COMPETENCE EVALUATIONS

4.1. Method

A correlational study was conducted amongst English nationals to test the predictive power of SDO alongside three control variables (age, gender and regional identity in England) upon the competence of Northern English and Southern English speech at explicit and implicit evaluational levels. Age, gender and regional identity were included as control variables since prior research findings indicated all three factors were especially robust predictors of English nationals' self-report and automatic language attitudes (see Bishop et al., 2005; Coupland and Bishop, 2007; McKenzie and Carrie, 2018; Sharma et al., 2022).

4.1.1. Participants

Three hundred and seven participants partook in an online study via Prolific (male = 161, female = 143, other gender = 2, missing = 1). All participants self-identified as English nationals. Participant ages ranged from 18 to 78 ($M = 43.66$, $SD = 15.98$). Regarding regional identification, 129 self-identified as 'northern English', 133 as 'southern English', 44 as 'other' (English), with 1 missing value. Given the small participant numbers, responses pertaining to the 'other' gender were excluded from the gender analysis. The 'other' regional identity (i.e., neither northern English nor southern English) responses were also excluded from the regional identity analysis.

4.1.2. Explicit attitude measure

Explicit competence evaluations were assessed through two self-report 100-point magnitude estimation scales (1 = lowest, 100 = highest). The items on each scale incorporated five evaluative traits obtained through prior language research involving English nationals (McKenzie and Carrie, 2018): (i) correct; (ii) high-status; (iii) clear; (iv) educated; and (v) confident. There were 10 items in total: five consisting of 'The varieties of English spoken in the north of England sound...' and a further five consisting of 'The varieties of English spoken in the south of England sound...', each of which included one of the five evaluative traits detailed above. The two instruments were thus identical except that one evaluational scale requested participants to rate speakers of varieties spoken in the north of England whilst the other requested they rate speakers of varieties spoken in the south of England. Analysis demonstrated high reliability for both the Northern English speech evaluation scale ($\alpha = 0.826$) and the Southern English speech evaluation scale ($\alpha = 0.773$).

4.1.3. Implicit attitude measure

To assess participants' implicit language attitudes an auditory Implicit Association Test (see Campbell-Kibler, 2012; Vari and Tamburelli, 2020) was specially constructed. The speech stimuli utilised consisted of digital recordings prototypical of the standard English forms employed in (i) northern England (i.e., General Northern English) (GNE) and (ii) southern England (i.e., Standard Southern British English) (SSBE) (e.g., Cruttenden, 2014); the most socially meaningful geographical distinction made between regional varieties in England (Wales, 2006; McKenzie and Carrie, 2018). Specifically, short tokens of the BATH vowel uttered in a series of one syllable words – where English speakers from the north employ the shorter [a] and speakers from the south of England use the longer [ɑ:] in lexical items such as 'bath', 'grass' and 'laugh' – were presented as the most marked variants of Northern English speech and Southern English speech respectively (Wells, 1982; Docherty et al., 2024). To control for potential speaker gender, age and educational level confounds, the GNE and SSBE tokens were provided by two females in their early thirties who were both educated to PhD level. The prototypicality of the BATH speech tokens as either Northern English speech or Southern English speech was validated in a pre-test stage involving English nationals listening to and categorising the regional provenance of both speakers (see also McKenzie, 2015). The same five positive adjectives utilised in the explicit competence scale, together with their binary opposites, were also employed within the competence IAT.

Table 1
Implicit association test (IAT) blocks.

Block	Trials	Type	Stimuli
1	20	Practice	Adjectives
2	20	Practice	Audio
3	40	Experimental	Adjectives + Audio
4	20	Practice	Audio reversed association
5	80	Experimental	Adjectives + Audio reversed association

Following Greenwald et al. (2003), five blocks were incorporated into the IAT design (Table 1): Blocks 1, 2 and 4 constituted the practice tasks whilst Blocks 3 and 5 represented the experimental tasks. In accordance with Greenwald et al. (ibid), to control for the reversing of the audio association, more trials were employed within experimental Block 5 (80) than Block 3 (40).

During the IAT task, in-line with established procedure, participants were instructed to utilise either the left or right arrow key on their respective keyboard to classify each adjective which appeared on the screen as either positive or negative in meaning. Participants were likewise requested to categorise each audio sample presented as either Southern English speech or Northern English speech. The order of the experiment was counter-balanced whereby participants commenced the IAT blocks with either the left-positive or right-positive pairings.

4.1.4. Social dominance orientation measure

SDO was assessed by means of the 16-item SDO scale developed by Pratto et al. (1994). Participants recorded their strength of agreement or disagreement on a 100-point magnitude estimation scale. Following Ho et al. (2012), the scale consisted of an equal number of 8 protrait, support for group-based hierarchy items, together with 8 contrait, opposition to group-based hierarchy items (see Appendix). Analysis indicated high reliability for the overall scale ($\alpha = 0.95$).

4.1.5. Demographic information

In light of evidence obtained from prior language attitude research detailing individual differences in listener evaluations, including in England, participants were requested to provide information regarding their age, gender, country of birth, nationality, native language and regional identification as 'northern English', 'southern English' or 'other' (English). The latter set of participant responses constituted the *regional identity* (in England) variable.

4.1.6. Procedure

Participants were recruited through the online recruitment platform, Prolific, and received a small monetary payment to partake. Only those individuals aged 18 or over, who registered their nationality as English and were resident in England during the data collection period were invited. Following completion of the explicit, SDO and demographic measures on Qualtrics, participants were automatically redirected to the IAT on Pavlovio. The online data collection method enabled the recruitment of a large cohort of adult English nationals in stages – allowing for the targeting of particular participants at each stage – and ensuring a balanced sample in terms of age, gender, and regional affiliation in England.

4.2. Results

4.2.1. Descriptive statistics

The initial analysis, prior to the examination of any SDO effect, involved the calculation of the English participants' responses to Southern English and Northern English speech on the explicit attitude and implicit attitude measures. Explicit competence evaluations of the Southern English speech were generally positive ($M = 71.77$, $SD = 13.63$) (where 1 = least positive and 100 = most positive). Conversely, explicit competence evaluations of Northern English speech were significantly less favourable ($M = 56.04$, $SD = 17.72$) ($t(305) = 13.31$, $p < 0.01$). Both sets of explicit evaluations were normally distributed.

The IAT competence scores were calculated in accordance with Greenwald et al. (2003). First, responses greater than 10 s in the two experimental trials were excluded from the analysis. Participants who provided more than 10% of their responses under 300 ms were also excluded. The remaining responses were used to calculate an overall D score effect size. D scores are bounded by -2 and $+2$. In the current research, a positive competence score represented an implicit bias towards Southern English speech whereas a negative score represented an implicit bias towards Northern English speech. The D-IAT score indicated a significant implicit competence bias towards Southern English speech ($M = 0.22$, $SD = 0.58$) ($t(305) = 6.80$, $p < 0.01$).

Initial analysis of the SDO data involved the scoring of participants' protrait responses and the reverse scoring of their contrait responses. The eight protrait and eight contrait item scores were subsequently aggregated to form an overall SDO score on the 100-point magnitude estimation scale. The analysis revealed somewhat low overall levels of SDO ($M = 19.78$, $SD = 18.40$) (where 1 = lowest and 100 = highest SDO).

4.2.2. Bivariate correlations

To investigate SDO as a potential determinant of self-report and automatic competence evaluations, correlations between each of the dependent and independent variables were calculated. A significant negative correlation was demonstrated between SDO and explicit competence evaluations of Northern English speech ($r = -0.22$, $p < 0.01$).

(Table 2). Conversely, no correlation between SDO and Southern English competence evaluations was found. Likewise, no relationship was uncovered between SDO and implicit competence evaluations of Northern English speech-Southern English speech (calculated as a single binary score).

Table 2

Mean scores and bivariate correlations (Pearson's r) for all variables across Study 1 (Competence).

Variable	<i>N</i>	Mean (<i>SD</i>)	Age	Gender	Regional Identity	SDO	Implicit Comp	Explicit Comp South Eng	Explicit Comp North Eng
Age	306	43.66 (15.98)	—						
Gender	304	0.47 (0.5)	0.028	—					
Regional Identity	262	0.51 (0.5)	−0.013	0.096	—				
SDO	304	19.78 (18.4)	0.068	−0.135*	0.071	—			
Implicit Comp	306	0.22 (0.58)	0.238***	0.139*	0.419***	0.063	—		
Explicit Comp South Eng	306	71.77 (13.63)	0.007	0.112	0.324***	−0.063	0.144*	—	
Explicit Comp North Eng	306	56.04 (17.72)	0.065	0.079	−0.232***	−0.221***	−0.211***	0.149**	—

Note. Significant correlations presented in boldface.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; gender ($F = 1$, $M = 0$), regional identity (southern = 1, northern = 0).

4.2.3. Regression analysis of SDO upon explicit competence evaluations

Hierarchical multiple regression analysis was conducted to control for the effect of listener age, gender (dummy coded, male = 0, female = 1), regional identity (dummy coded, northern English affiliation = 1, southern English affiliation = 2) and age (continuous variable) upon participants' explicit and implicit competence evaluations. Specifically, participants' categorisations of these three variables were incorporated to form an initial Model 1. SDO responses were subsequently added to form Model 2 to assess how much additional variance, if any, SDO explains in relation to the implicit and explicit competence evaluations.

As detailed in Table 3, Model 1, composed solely of the age, gender and regional identity, accounted for 7% of the variance ($R^2 = 0.070$, $F(3,258) = 6.424$, $p < 0.001$) of the explicit competence evaluations of Northern English speech. The incorporation of SDO into Model 2 significantly increased the variance explained to 12% ($R^2 = 0.12$, $F(4, 258) = 8.641$, $p < 0.001$), (R^2 change = 0.042, F change (1, 258) = 10.357, $p < 0.01$). The analysis indicated a one-point increase in SDO was linked to a reduction in explicit Northern English speech competence evaluations of 0.23.

Table 3

Regression coefficients for the effect of SDO upon explicit competence evaluations of Northern English speech.

Model	R^2		<i>B</i>	<i>SE</i>	β	<i>p</i>
1	0.070	Age	0.091	0.068	0.082	0.178
		Gender	3.031	2.173		0.164
		Regional Identity	−8.74	2.172		<.001
2	0.120	SDO	−0.235	0.062	−0.227	<.001

Table 4 indicates SDO did not predict explicit Southern English speech competence evaluations. Specifically, whilst Model 1, incorporating the three control variables, was significant ($R^2 = 0.113$, $F(3,258) = 10.832$, $p < 0.001$), the inclusion of SDO within Model 2 did not significantly increase the predictive value ($R^2 = 0.114$, $F(4,258) = 8.149$, $p = 0.654$).

Table 4

Regression coefficients for the effect of SDO upon explicit competence evaluations of Southern English speech.

Model	R^2		<i>B</i>	<i>SE</i>	β	<i>p</i>
1	0.113	Age	0.051	0.049	0.061	0.304
		Gender	2.084	1.578		0.188
		Regional Identity	8.38	1.577		<.001
2	0.114	SDO	−0.021	0.046	−0.027	0.654

4.2.4. Regression analysis of SDO upon implicit competence evaluations

As detailed in Table 5, Model 1 was found to be significant ($R^2 = 0.252$, $F(3,258) = 28.697$, $p < 0.001$). However, the incorporation of the SDO into Model 2 did not increase the predictive value ($R^2 = 0.259$, $F(4,258) = 22.249$, $p = 0.121$).

Table 5

Regression coefficients for the effect of SDO upon implicit competence evaluations.

Model	R^2		B	SE	β	p
1	0.252	Age	0.009	0.002	0.239	<.001
		Gender	0.12	0.063		0.056
		Regional Identity	0.484	0.063		<.001
2	0.259	SDO	0.003	0.002	0.086	0.121

5. STUDY 2: WARMTH EVALUATIONS

5.1. Method

A correlational design was employed to assess the predictive power of SDO upon the English nationals' explicit and implicit warmth evaluations of Northern English and Southern English speech.

5.1.1. Participants

The same cohort of participants who completed Study 1 were invited to partake in Study 2. A slightly smaller subset of these English nationals volunteered ($N = 240$) (male = 130, female = 108, other gender = 2). Participant ages ranged from 18 to 78 ($M = 46.18$, $SD = 15.64$). Ninety-six participants self-identified as 'northern English', 107 as 'southern English' and 37 as 'other' (English). Given the small participant numbers, the responses of the 'other gender' and 'other English' groups were again excluded from the analysis.

5.1.2. Measures

With one key exception, the two explicit measures employed in Study 2 were identical to the measures utilised in Study 1: the five competence adjectives were replaced by the following *warmth* adjectives: i) pleasant; ii) friendly; iii) good; v) honest; and v) sociable. High reliability was again demonstrated for both the Northern English speech ($\alpha = 0.890$) and the Southern English speech ($\alpha = 0.917$) evaluational scales. The auditory IAT also largely replicated the Study 1 design. Again, the sole difference between the two IATs related to the evaluative traits employed, i.e., the competence evaluative traits employed within Study 1 were replaced by warmth traits (together with their bipolar opposites) in Study 2. The data obtained from the 16-item SDO scale and the demographic questionnaire utilised in Study 1 was again employed in Study 2.

5.1.3. Procedure

Study 2 was conducted several weeks after Study 1 to avoid participant fatigue and ensure independence of measurements. Participants who completed Study 2 were paid a further small sum. All participants initially undertook the explicit attitude task followed by the IAT task.

5.2. Results

5.2.1. Descriptive statistics

Whilst explicit warmth evaluations of both speech forms were broadly favourable, contrary to the competence evaluations obtained, participants were significantly more positive towards Northern English speech ($M = 71.00$, $SD = 17.82$) than Southern English speech ($M = 63.18$, $SD = 18.22$) (where 1 = least positive and 100 = most positive) ($t(239) = 5.31$, $p < 0.01$). The D-IAT score, conversely, indicated an implicit warmth bias towards Southern English speech ($M = 0.12$, $SD = 0.44$) ($t(239) = 3.32$, $p < 0.01$), mirroring the competence bias found in Study 1.

5.2.2. Bivariate correlations

Preliminary correlational analysis was undertaken to assess the effect of SDO upon the participants' self-report and automatic warmth evaluations of Northern English and Southern English speech. Table 6 shows that whilst a significant negative correlation was found between SDO and explicit warmth evaluations of Northern English speech ($r = -0.22$, $p < 0.01$), no correlation was uncovered between SDO and explicit warmth Southern English speech evaluations. Analysis showed a significant positive correlation between SDO and implicit warmth evaluations of Southern English ($r = 0.16$, $p = 0.01$) (and thus, correspondingly, a negative association between SDO and implicit warmth evaluations of Northern English speech).

Table 6
Mean scores and bivariate correlations (Pearson's r) for all variables across Study 2 (Warmth).

Variable	N	Mean (SD)	Age	Gender	Regional Identity	SDO	Implicit Warm	Explicit Warm South Eng	Explicit Warm North Eng
Age	240	46.18 (15.64)	—						
Gender	238	0.47 (0.5)	0.028	—					
Regional Identity	203	0.51 (0.5)	-0.013	0.096	—				
SDO	240	19.78 (18.40)	0.068	-0.135*	0.071	—			
Implicit Warm	231	0.12 (0.44)	0.163*	0.149*	0.489***	0.113*	—		
Explicit Warm South Eng	240	63.18 (18.22)	0.018	0.232***	0.216**	-0.099	0.277***	—	
Explicit Warm North Eng	240	71.00 (17.82)	0.099	0.087	-0.286***	-0.227***	-0.273***	0.228***	—

Note. Significant correlations presented in boldface.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; gender (F = 1, M = 0), regional identity (southern = 1, northern = 0).

5.2.3. Regression analysis of SDO upon explicit warmth evaluations

Age, gender and regional identity were again employed as control variables to form Model 1. SDO was again incorporated alongside the three control variables to form Model 2.

Table 7 shows Model 1 accounted for 10.5% of the variance ($R^2 = 0.105$, $F(3,199) = 7.626$, $p < 0.001$). The incorporation of SDO into Model 2 significantly increased the variance explained to 15.1% ($R^2 = 0.151$, $F(4,199) = 8.691$, $p < 0.0001$) (R^2 change = 0.048, F change (1,199) = 11.078, $p < 0.001$), and indicated a one-point increase in SDO was linked to a reduction in explicit Northern English speech warmth evaluations of 0.22.

Table 7
Regression coefficients for the effect of SDO upon explicit warmth evaluations of Northern English speech.

Model	R^2		B	SE	β	p
1	0.105	Age	0.114	0.078	0.099	0.145
		Gender	3.696	2.447		0.132
		Regional Identity	-10.362	2.454		<.001
		SDO	-0.219	0.067		0.001
2	0.151				-0.22	0.001

Table 8 indicates that Model 1 was significant ($R^2 = 0.108$, $F(3,199) = 7.928$, $p < 0.001$). However, the addition of the SDO into Model 2 did not increase the predictive value ($R^2 = 0.110$, $F(4,199) = 6.019$, $p = 0.543$).

Table 8
Regression coefficients for the effect of SDO upon explicit warmth evaluations of Southern English speech.

Model	R^2		B	SE	β	p
1	0.108	Age	0.062	0.08	0.053	0.439
		Gender	9.104	2.517		<.001
		Regional Identity	7.554	2.524		0.003
		SDO	-0.043	0.07		0.543
2	0.110				-0.042	0.543

5.2.4. Regression analysis of SDO upon implicit warmth evaluations

Table 9 indicates Model 1 was significant ($R^2 = 0.276$, $F(3,192) = 23.976$, $p < 0.001$). Again, the inclusion of SDO in Model 2 did not produce a significant change in predictive value ($R^2 = 0.288$, $F(4,192) = 18.975$, $p = 0.077$).

Table 9

Regression coefficients for the effect of SDO on implicit warmth evaluations.

Model	R^2		B	SE	β	p
1	0.276	Age	0.006	0.002	0.179	0.004
		Gender	0.097	0.067		0.146
		Regional Identity	0.529	0.067		<.001
		SDO	0.003	0.002		0.077
2	0.288				0.112	

6. SUMMARY OF EFFECTS: STUDY 1 (COMPETENCE) AND STUDY 2 (WARMTH)

Table 10 details the significant effects, reported as standardised beta weights, for both Study 1 (Competence) and Study 2 (Warmth).

Table 10

Summary of statistically significant findings: The effect of SDO upon explicit and implicit evaluations of Northern English speech and Southern English speech.

	Northern English speech		Southern English speech		Northern English-Southern English speech	Northern English-Southern English speech
	Explicit Competence	Explicit Warmth	Explicit Competence	Explicit Warmth	Implicit Competence (one score)	Implicit Warmth (one score)
SDO	-0.23	-0.22				

7. GENERAL DISCUSSION AND CONCLUSIONS

The preliminary stage of the present research examined English participants' explicit and implicit competence and warmth evaluations of two forms of English spoken in England: Northern English and Southern English speech. Consistent with the compensatory competence-warmth evaluations uncovered within prior language attitude research undertaken in England employing self-report measures, analysis indicated that whilst English participants judged Southern English speech more positively for competence, they evaluated Northern English speech more highly for warmth. However, analysis of the IAT responses indicated an overall implicit bias in favour of Southern English speech and, correspondingly, against Northern English speech for both competence and warmth. This result provides evidence of English nationals' explicit status-related prejudice towards Northern English speech forms and by extension, the community of speakers, together with greater implicit bias against (speakers of) Northern English for both warmth and competence. Intriguingly, implicit prejudice was uncovered even when General Northern English (GNE), the most standard English speech form employed in the north of England (Cruttenden, 2014; Strycharczuk et al., 2020; Dann et al., 2024), was presented for evaluation; suggesting, for English nationals, Standard Southern British English (SSBE) functions as a high-status national standard in England whilst GNE functions as a comparatively lower status regional standard in the north of England only.

The primary objective of the current research, nonetheless, was to assess the effect of English nationals' SDO levels – indicative of preference for inequality between social groups – upon their explicit and implicit language attitudes. Age, gender and regional identity were controlled for the purposes of the hierarchical regression analysis of the participants' responses and a detailed discussion of the results of this regression analysis, as opposed to the solely preliminary bivariate correlation analysis, is provided below with regards to the specific hypotheses formulated.

Hypothesis 1a stated that English nationals with high-SDO would hold more positive explicit competence, but not warmth, attitudes towards Southern English speech. *Hypothesis 1a* was only partly confirmed since, when controlling for age, gender and regional identity, no effect for SDO was found upon either the competence or warmth self-report

evaluations of Southern English. Since the SDO score was obtained through participants' explicit evaluations, and SDO is generally considered to predict competence-incompetence judgments (Sibley et al., 2007), the absence of an effect upon explicit competence judgements of standard forms of Southern English speech – frequently afforded high-status by English nationals in prior language attitude studies – may seem surprising.

A significant SDO effect, however, was demonstrated upon the explicit competence evaluations of Northern English speech: specifically, high-SDO participants were significantly less positive when compared to low-SDO participants. This result parallels the general pattern uncovered in prior studies investigating self-report attitudes towards non-language-related stimuli, where evidence exists, that when two social groups were evaluated together, SDO was a stronger determinant of competence attitudes towards the perceived subordinate group than the perceived dominant group (Rowatt et al., 2005; Stern and Axt, 2019).

Given considerable evidence demonstrating listeners' language attitudes reflect evaluations of the perceived community of speakers more broadly (McKenzie et al., 2019; Dragojevic et al., *in press*) the results of the present research point to SDO as primarily a predictor of *explicit* linguistic prejudice against Northern English speech *precisely because* the north of England, and its inhabitants, are perceived to be lower in status and subordinate by English nationals when compared with the inhabitants from the politically, economically and socially more dominant south of the country. Similar findings were found within the few prior explicit language attitude studies examining SDO as an individual difference variable, where SDO was found to predict negative self-report attitudes towards comparatively lower status speakers of Latino English (Hansen and Dovidio, 2016) and L2 English speech (Sumantry and Choma, 2021) in the United States as well as towards a perceived subordinate group of non-native Polish speakers in Poland (Hansen, 2020).

Our analysis further indicated that high-SDO predicted negative self-report warmth attitudes towards Northern English, i.e., participants with low-SDO typically evaluated Northern English speech significantly more favourably. *Hypothesis 1b*, which stated that high-SDO English nationals would hold more negative explicit competence, but not warmth, attitudes towards Northern English speech, was thus only partially confirmed.

Taken together, the results indicated that high-SDO amongst English nationals can determine especially intense levels of explicit competence-related linguistic biases, and thus language-based prejudices, against speakers perceived as subordinate as opposed to predicting linguistic favouritism towards dominant groups of speakers. The negative association found between SDO and warmth-related self-report evaluations – where low-SDO English nationals evaluated the Northern English speech more highly in terms of warmth – provides intriguing evidence that SDO, generally considered a status-endorsing variable, can also determine warmth explicit evaluations of subordinate communities of speakers.

With regards to the influence of SDO on more deeply-embedded evaluations, *Hypothesis 2* stated that English nationals with high-SDO would hold more negative implicit competence, but not warmth, biases against Northern English than Southern English speech. Regression analysis indicated that, when compared to low-SDO participants, high-SDO participants evaluated Southern English most positively and, accordingly, Northern English speech more negatively for both competence and warmth. Whilst preliminary analysis found a correlation between SDO and implicit warmth, more fine-grained hierarchical multiple regression analysis – controlling for age, gender and regional identity – found no significant effect for SDO upon either the implicit competence or warmth biases against Northern English speech. Consequently, *Hypothesis 2* was not confirmed.

The presence of an SDO effect upon explicit attitudes, and the absence of an effect upon implicit attitudes, towards Northern English and Southern English speech parallels the findings of several other studies assessing the effect of SDO upon evaluations of non-language based stimuli, whereby high-SDO was associated with explicit but not implicit competence-related prejudice towards groups perceived to be subordinate in terms of religion (Rowatt et al., 2005), sexuality (Anderson and Koc, 2015; Moor et al., 2019) and towards evaluations of Barack Obama and foreignness (Crawford and Bhatia, 2012).

The possibility exists that the absence of a significant effect for SDO upon implicit responses was due to the specific measures developed in the current research. For instance, it cannot be discounted that the employment of two IATs, which typically generate relatively small effect sizes from participant responses (Oswald et al., 2015), may have resulted in less sensitive implicit attitude measures – and hence minimised any potential effect of differences in SDO responses – when compared to the magnitude estimation scales employed to capture explicit attitudes.

A more plausible explanation is that SDO functions at explicit levels and, consequently, is a more robust determinant of self-report as opposed to automatic attitudes. This may be because researchers have generally obtained SDO scores by means of self-report instruments; indeed, it seems that implicit measures of SDO have yet to be developed. From this perspective, it is unsurprising that SDO was found to predict explicit language attitudes only and helps explain why few prior studies indicated SDO predicted implicit attitudes towards other social categories.

In summary, this is the first research to examine the effect of SDO upon *both* automatic and self-report evaluations of linguistic stimuli. Hierarchical multiple regression analysis indicated that SDO level determined English nationals' explicit

but not implicit language attitudes. Specifically, regression analyses, controlling for age, gender and regional identity, substantiated SDO as a predictor of self-report evaluations of the perceived *subordinate speech community* for both warmth and competence. This result extends prior linguistic research conducted by Hansen and Dovidio (2016) and Hansen (2020) which found an effect for high-SDO upon negative competence explicit evaluations but did not examine the role of SDO on warmth attitudes. The significant effect for high-SDO upon on English nationals' negative self-report warmth evaluations of Northern English demonstrated in the current research also provides contradictory evidence to Sumantry and Choma's (2021) investigation of explicit attitudes towards L1 and L2 English speech amongst US nationals where, although high-SDO was found to be a predictor of listeners' negative competence evaluations, no effect was demonstrated for SDO for warmth. To validate (or not) the negative effect for high-SDO upon warmth evaluations of linguistic varieties and their speakers, there thus seems a specific requirement to undertake further language attitude research incorporating both warmth and competence measures into the study design as well as to examine the ways in which SDO may influence evaluations of other speech communities outwith England and the United States.

The implicit-explicit attitudinal discrepancy (IED) (Ianos et al., 2020; McKenzie and Carrie, 2018) revealed by the differences in the English nationals' SDO responses in the current research also adds to the findings of numerous prior empirical studies indicating that implicit and explicit attitudes are distinct constructs which operate at different evaluational levels (Petty et al., 2009). Given the plethora of existing studies investigating the influence of SDO upon self-report and/or automatic evaluations of other social groups, the results of the current research point to the theoretical and methodological value for language scientists to undertake additional empirical research specifically investigating the effect of SDO, in conjunction with other individual differences, upon both implicit and explicit attitudes towards linguistic stimuli. The results obtained from future studies may, for instance, offer valuable information regarding the direction of any language attitude change in progress, most probably at explicit level of evaluation (McKenzie and Carrie, 2018; Charlesworth and Banaji, 2019), together with insights into the groups of individuals – for instance those higher or lower in social dominance orientation – who lead, or are most resistant to, attitude change towards specific speech communities and the linguistic forms they employ.

Nevertheless, the investigation of the role of SDO upon implicit and explicit language attitudes remains in its infancy. The results of the present research should be viewed as exploratory and there are several limitations regarding the study design. For instance, there exists a potential lack of validity comparing English nationals' responses to the different forms of stimuli presented for evaluation within the explicit and implicit measures. Specifically, it is possible the incorporation of the conceptual labels 'Southern English' and 'Northern English' as stimuli within the self-report magnitude estimation measures and, conversely, the employment of Standard Southern British English (SSBE) alongside General Northern English (GNE) speech tokens of the BATH variable as evaluative stimuli within the IAT measures may, to some extent, be responsible for the discrepancy between the SDO effects upon participants' explicit and implicit language attitudes. To test for this potential confound, future equivalent research could include the same speech stimuli within the self-report and automatic measures. Likewise, since the IAT measures employed female speech as stimuli, to validate (or not) the results of the present project, future research could incorporate tokens of male speech stimuli within the study design.

In order to minimise the cognitive load for participants, a deliberate decision was taken not to combine competence and warmth evaluative traits within the same measures (see Jacard, 2025). Instead, separate studies were run, consisting of explicit and implicit attitude measures incorporating competence traits (Study 1) or warmth traits (Study 2). It is worth noting, however, that a slightly lower number of the participants took part in the warmth study when compared with the competence study and thus, it cannot be discounted that the difference in sample size may, to some extent, have contributed to the difference between English nationals' competence and warmth responses to the two varieties of English.

Relatedly, the specific employment of two implicit association tests to gauge the English nationals' implicit competence and warmth evaluations in the current study is potentially problematic since, by its very design, the IAT typically uncovers small differences between participants' response latencies – albeit with socially important implications – with regards to the social categories under investigation (Greenwald et al., 2015). In order to help validate (or not) the implicit findings gained in the present study, it would be worthwhile for researchers investigating the potential effect of SDO upon implicit language-based evaluations to employ other implicit attitude measures, e.g., the Go/No-Go Associative Task (Nosek and Banaji, 2001), the Affect Misattribution procedure (AMP) (Payne et al., 2005) or the Relational Responding Task (De Houwer et al., 2015).

Researchers have also recently questioned whether SDO functions as a unitary construct. Some posit that SDO is composed of two related sub-dimensions: i) SDO-D, associated with more covert, old-fashioned prejudice; and ii) SDO-E, associated with more subtle, modern prejudice (Ho et al., 2012). Empirical research has indicated that SDO-D and SDO-E may indeed predict different components of prejudicial self-report attitudes towards political (Bergh et al., 2015) and ethnic groups (Jost and Thompson 2000). However, the extent to which, if at all, SDO-D and/or SDO-E relate to

self-report and automatic evaluations of language varieties and their communities of speakers is currently unknown and further sociolinguistic research incorporating both these SDO sub-dimensions seems necessary.

A deliberate decision was made to assess the influence of SDO and to control for age, gender and regional affiliation, previously found to determine self-report and/or automatic language attitudes. However, regression analysis indicated these predictor variables together accounted for only a relatively small amount of the variance within English nationals' implicit and explicit competence and warmth responses: between 7% and 15%. To help account for other variance within the participants' explicit and implicit judgments, there is a requirement for further studies exploring the role of SDO in conjunction with other individual differences previously found to predict listeners' linguistic evaluations; for example, ethnicity, educational background, socio-economic status, personality or processing fluency (McKenzie, 2010; Dragojevic and Giles, 2016).

Similarly, there are undoubtedly other individual difference variables, seldom examined by linguists, which could be incorporated from socio-psychological research and investigated as potential determinants of explicit and implicit language attitudes. For instance, psychologists have found right-wing authoritarianism (RWA) – an ideological construct constituting an individual's level of authoritarian aggression, authoritarian submission and conventionalism (Altemeyer, 1996) – predicted explicit and implicit attitudes towards a range of social categories (Duckitt and Sibley, 2009). Indeed, SOD and RWA are often examined within the same attitude study and research – especially in western European contexts – has demonstrated significant correlations between the two individual differences (see Duckitt and Sibley, 2007). Nonetheless, researchers have yet to establish the precise role, if any, which RWA – and other hierarchically-structured individual differences examined within social psychology, e.g. system justification (Jost, 2020) – in conjunction with SDO may play in determining language attitudes.

FUNDING

This research project was supported by a British Academy Mid-Career Fellowship [grant number MD20/200009].

DECLARATION OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Robert M. McKenzie: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kristofor McCarty:** Software. **Mimi Huang:** Writing – review & editing.

Data availability

Data will be made available on request.

APPENDIX

Items comprising the SDO measure

Protrait items

Some groups of people are simply inferior to other groups

In getting what you want, it is sometimes necessary to use force against other groups

It's OK if some groups have more of a chance in life than others

To get ahead in life, it is sometimes necessary to step on other groups

If certain groups stayed in their place, we would have fewer problems

It's probably a good thing that certain groups are at the top and other groups are at the bottom

Inferior groups should stay in their place

Sometimes other groups must be kept in their place

Contrait items

It would be good if groups could be equal

Group equality should be our ideal
 All groups should be given an equal chance in life
 We should do what we can to equalize conditions for different groups
 Increased social equality is beneficial to society
 We would have fewer problems if we treated people more equally
 We should strive to make incomes as equal as possible
 No group should dominate in society

References

- Abele, A.E., Ellemers, N., Fiske, S.T., Koch, A., Yzerbyt, V., 2021. Navigating the social world: toward an integrated framework for evaluating self, individuals, and groups. *Psychol. Rev.* 128 (2), 290–314. <https://doi.org/10.1037/rev0000262>.
- Altemeyer, B., 1996. *The Authoritarian Specter*. Harvard University Press, Cambridge, Mass.
- Álvarez-Mosquera, P., Marín-Gutiérrez, A., 2021. Language attitudes in a lingua franca: the case of black South African college students. *J. Lang. Soc. Psychol.* 40 (3), 378–397. <https://doi.org/10.1177/0261927X20951904>.
- Anderson, J., 2018. Implicit and explicit attitudes towards asylum seekers in Australia: demographic and ideological correlates. *Aust. Psychol.* 53 (2), 181–191. <https://doi.org/10.1111/ap.12229>.
- Anderson, J., Koc, Y., 2015. Exploring patterns of explicit and implicit anti-gay attitudes in muslims and atheists. *Eur. J. Soc. Psychol.* 45 (6), 687–701. <https://doi.org/10.1002/ejsp.2126>.
- Baker, C., 1992. *Attitudes and Language*. Multilingual Matters, Clevedon.
- Bergh, R., Sidanius, J., Sibley, C.G., 2015. Dimensions of social dominance: their personality and socio-political correlates within a New Zealand probability sample. *N. Z. J. Psychol.* 44 (2), 25–34.
- Bishop, H., Coupland, N., Garrett, P., 2005. Conceptual accent evaluation: thirty years of accent prejudice in the UK. *Acta Linguist. Hafniensia* 37 (1), 131–154. <https://doi.org/10.1080/03740463.2005.10416087>.
- Braber, N., Smith, H., Wright, D., Hardy, A., 2023. Assessing the specificity and accuracy of accent judgements by lay listeners. *Lang. Speech* 66 (2), 267–290. <https://doi.org/10.1177/00238309221101560>.
- Calamai, S., Ardolino, F., 2020. Italian with an accent: the case of “Chinese Italian” in Tuscan high schools. *J. Lang. Soc. Psychol.* 39 (1), 132–147. <https://doi.org/10.1177/0261927X19883899>.
- Campbell-Kibler, K., 2012. The implicit association test and sociolinguistic meaning. *Lingua* 22 (7), 753–763. <https://doi.org/10.1016/j.lingua.2012.01.002>.
- Charlesworth, T.E.S., M.R. Banaji, M.R., 2019. Patterns of implicit and explicit attitudes: I. Long-term change and stability from 2007 to 2016. *Psychol. Sci.* 30(2), 174–192. doi: 10.1177/0956797618813087.
- Coupland, N., Bishop, H., 2007. Ideologised values for British accents. *J. Socioling.* 11 (1), 74–93. <https://doi.org/10.1111/j.1467-9841.2007.00311.x>.
- Crawford, J.T., Bhatia, A., 2012. Birther nation: political conservatism is associated with explicit and implicit beliefs that President Barack Obama is foreign: line conservatism and Obama’s foreignness. *Anal. Soc. Issues Public Policy* 12 (1), 364–376. <https://doi.org/10.1111/j.1530-2415.2011.01279.x>.
- Cruttenden, A., 2014. *Gimson’s Pronunciation of English*, eighth ed. Routledge, London.
- Dann, H., Drummond, R., Tasker, S., Montgomery, C., Ryan, S.D., Carrie, E., 2024. Broad, strong, and soft: using geospatial analysis to understand folk-linguistic terminology. *J. Linguist. Geogr.* 12 (1), 44–56. <https://doi.org/10.1017/jlg.2024.10>.
- Dekker, S., Duarte, J., Loerts, H., 2021. ‘Who really speaks like that?’ – children’s implicit and explicit attitudes towards multilingual speakers of Dutch. *Int. J. Multiling.* 18 (4), 551–569. <https://doi.org/10.1080/14790718.2021.1908297>.
- De Houwer, J., Heider, N., Spruyt, A., Roets, A., Hughes, S., 2015. The relational responding task: toward a new implicit measure of beliefs. *Front. Psychol.* 6, 00319. <https://doi.org/10.3389/fpsyg.2015.00319>.
- De Houwer, J., Moors, A., 2007. How to define and examine the implicitness of implicit measures. In: Wittenbrink, B., Schwarz, N. (Eds.), *Implicit Measures of Attitudes*. The Guilford Press, New York, pp. 179–194.
- Docherty, G., Foulkes, P., Kerswill, P., 2024. Phonetic and phonological variation in England. In: Fox, S. (Ed.), *Language in Britain and Ireland*. third ed. Cambridge University Press, Cambridge, pp. 70–97.
- Dragojevic, M., Berglund, C., Blauvelt, T.K., 2015. Attitudes toward Tbilisi - and Mingrelian-accented Georgian among Georgian youth: on the road to linguistic homogenization? *J. Lang. Soc. Psychol.* 34 (1), 90–101. <https://doi.org/10.1177/0261927X14555191>.
- Dragojevic, M., Giles, H., 2016. I don’t like you because you’re hard to understand: the role of processing fluency in the language attitudes process. *Hum. Commun. Res.* 42 (3), 396–420. <https://doi.org/10.1111/hcre.12079>.
- Dragojevic, M., Giles, H., Goatley-Soan, S., Dayton, Z.A., in press. Americans’ attitudes toward British accents: the role of social categorisation, perceived group prototypicality, and processing fluency. *J. Multiling. Multicul. Dev.*, 1–18. doi: 10.1080/01434632.2024.2346575.
- Duckitt, J., Sibley, C.G., 2007. Right wing authoritarianism, social dominance orientation and the dimensions of generalized prejudice. *Eur. J. Pers.* 21 (2), 113–130. <https://doi.org/10.1002/per.614>.

- Duckitt, J., Sibley, C.G., 2009. A dual-process motivational model of ideology, politics and prejudice. *Psychol. Inq.* 20 (2–3), 98–109. <https://doi.org/10.1080/10478400903028540>.
- Edwards, J., 2011. *Challenges in The Social Life of Language*. Palgrave Macmillan, Basingstoke.
- Fiske, S.T., Cuddy, A.J.C., Glick, P., 2007. Universal dimensions of social cognition: warmth and competence. *Trends Cogn. Sci.* 11 (2), 77–83. <https://doi.org/10.1016/j.tics.2006.11.005>.
- Fiske, S.T., Taylor, S.E., 2021. *Social Cognition*. Sage, London.
- Garrett, P., 2010. *Attitudes to Language*. Cambridge University Press, Cambridge.
- Gawronski, B., De Houwer, J., 2014. Implicit measures in social and personality psychology. In: Reis, H.T., Judd, C.M. (Eds.), *Handbook of Research Methods in Social and Personality Psychology*. second ed. Cambridge University Press, Cambridge, pp. 283–310.
- Greenwald, A.G., Banaji, M.R., Nosek, B.A., 2015. Statistically small effects of the implicit association test can have societally large effects. *J. Pers. Soc. Psychol.* 108 (4), 553–561. <https://doi.org/10.1037/pspa0000016>.
- Greenwald, A.G., Nosek, B.A., Banaji, M.R., 2003. Understanding and using the implicit association test: I. An improved scoring algorithm. *J. Pers. Soc. Psychol.* 85 (2), 197–216. <https://doi.org/10.1037/0022-3514.85.2.197>.
- Greenwald, A.G., McGhee, D.E., Schwartz, J.L.K., 1998. Measuring individual differences in implicit cognition: the implicit association test. *J. Pers. Soc. Psychol.* 74 (6), 1464–1480. <https://doi.org/10.1037/0022-3514.74.6.1464>.
- Hansen, K., 2020. Accent beliefs scale (ABS): scale development and validation. *J. Lang. Soc. Psychol.* 39 (1), 148–171. <https://doi.org/10.1177/0261927X19883903>.
- Hansen, K., Dovidio, J.F., 2016. Social dominance orientation, nonnative accents, and hiring recommendations. *Cult. Divers. Ethn. Minor. Psychol.* 22 (4), 544–551. <https://doi.org/10.1037/cdp0000101>.
- Hiraga, N., 2005. British attitudes towards six varieties of English in the USA and Britain. *World Englishes* 24 (3), 289–308. <https://doi.org/10.1111/j.0883-2919.2005.00411.x>.
- Ho, A.K., Sidanius, J., Pratto, F., Levin, S., Thomsen, L., Kteily, N., Sheehy-Skeffington, J., 2012. Social dominance orientation: revisiting the structure and function of a variable predicting social and political attitudes. *Pers. Soc. Psychol. Bull.* 38 (5), 583–606. <https://doi.org/10.1177/0146167211432765>.
- Ianos, M.A., Rusu, A., Huguet, A., Lapresta-Rey, C., 2023. Implicit language attitudes in Catalonia (Spain): investigating preferences for Catalan or Spanish using the implicit association test. *J. Multiling. Multicul. Dev.* 44 (3), 214–229. <https://doi.org/10.1080/01434632.2020.1806857>.
- Jacard, J., 2025. Methodological issues in the study of implicit attitudes. In: Krosnick, J.A., Stark, T.H., Scott, A.L. (Eds.), *Implicit Bias and Racism*. Cambridge University Press, Cambridge, pp. 425–455.
- Jost, J.T., 2020. *A theory of system justification*. Harvard University Press, Cambridge.
- Jost, J.T., Thompson, E.P., 2000. Group-based dominance and opposition to equality as independent predictors of self-esteem, ethnocentrism, and social policy attitudes among African Americans and European Americans. *J. Exp. Soc. Psychol.* 36 (3), 209–232. <https://doi.org/10.1006/jesp.1999.1403>.
- Kinzler, K.D., Shutts, K., DeJesus, J., Spelke, E.S., 2009. Accent trumps race in guiding children's social preferences. *Soc. Cogn.* 27 (4), 623–634. <https://doi.org/10.1521/soco.2009.27.4.623>.
- Kteily, N.S., Ho, A.K., Sidanius, J., 2012. Hierarchy in the mind: the predictive power of social dominance orientation across social contexts and domains. *J. Exp. Soc. Psychol.* 48 (2), 543–549. <https://doi.org/10.1016/j.jesp.2011.11.007>.
- Labov, W., 2001. *Principles of Linguistic Change, vol 2: Social Factors*. Blackwell, Oxford.
- Lane, K.A., Banaji, M.R., Nosek, B.A., Greenwald, A.G., 2007. Understanding and using the implicit association test IV: what we know (so far) about the method. In: Wittenbrink, B., Schwarz, N. (Eds.), *Implicit Measures of Attitudes*. The Guilford Press, New York, pp. 59–102.
- Levon, E., Sharma, D., Watt, D.J.L., Cardoso, A., Ye, Y., 2021. Accent bias and perceptions of professional competence in England. *J. Engl. Linguist.* 49 (4), 355–388. <https://doi.org/10.1177/00754242211046316>.
- McKenzie, R.M., 2008. Social factors and non-native attitudes towards varieties of spoken English: a Japanese case study. *Int. J. Appl. Ling.* 18 (1), 63–88. <https://doi.org/10.1111/j.1473-4192.2008.00179.x>.
- McKenzie, R.M., 2010. *The Social Psychology of English as a Global Language*. Springer, Dordrecht.
- McKenzie, R.M., 2015. UK university students' folk perceptions of spoken variation in English: the role of explicit and implicit attitudes. *Int. J. Sociol. Lang.* 236, 31–53. <https://doi.org/10.1515/ijsl-2015-0020>.
- McKenzie, R.M., Carrie, E., 2018. Implicit-explicit attitudinal discrepancy and the investigation of language attitude change in progress. *J. Multiling. Multicul. Dev.* 39 (9), 830–844. <https://doi.org/10.1080/01434632.2018.1445744>.
- McKenzie, R.M., Gilmore, A., 2017. The people who are out of 'right' English: Japanese university students' social evaluations of English language diversity and the internationalisation of Japanese higher education. *Int. J. Appl. Ling.* 27 (1), 152–175. <https://doi.org/10.1111/ijal.12110>.
- McKenzie, R.M., Kitikanan, P., Boriboon, P., 2016. The competence and warmth of Thai students' attitudes towards varieties of English: the effect of gender and perceptions of L1 diversity. *J. Multiling. Multicul. Dev.* 37 (6), 536–550. <https://doi.org/10.1080/01434632.2015.1083573>.
- McKenzie, R.M., Huang, M., Ong, T.T., Snodin, N., 2019. Socio-psychological salience and categorisation accuracy of speaker place of origin. *Lingua* 228. <https://doi.org/10.1016/j.lingua.2019.06.006>.
- Milroy, J., Milroy, L., 2012. *Authority in Language*. Routledge, London.

- Moor, L., Kapelles, T., Koc, Y., Anderson, J., 2019. Predicting explicit and implicit attitudes towards gay men using the dual process model of prejudice and the dark tetrad. *Pers. Individ. Dif.* 151, <https://doi.org/10.1016/j.paid.2019.06.029> 109486.
- Nosek, B.A., Banaji, M.R., 2001. The go/no-go association task (GNAT). *Soc. Cogn.* 19 (6), 625–666. <https://doi.org/10.1521/soco.19.6.625.20886>.
- Nosek, B.A., Greenwald, A.G., Banaji, M.R., 2007. The implicit association test at age 7: A methodological and conceptual Review. In: Bargh, J.A. (Ed.), *Social Psychology and the Unconscious: the Automaticity of Higher Mental Processes*. Psychology Press, Hove, pp. 265–292.
- Oswald, F.L., Mitchell, G., Blanton, H., Jaccard, J., Tetlock, P.E., 2015. Using the IAT to predict ethnic and racial discrimination: small effect sizes of unknown societal significance. *J. Pers. Soc. Psychol.* 108 (4), 562–571. <https://doi.org/10.1037/pspa0000023>.
- Payne, B.K., Cheng, C.M., Govorun, O., Stewart, B.D., 2005. An inkblot for attitudes: affect misattribution as implicit measurement. *J. Pers. Soc. Psychol.* 89 (3), 277–293. <https://doi.org/10.1037/0022-3514.89.3.277>.
- Petty, R.E., Fazio, R.H., Brinol, P., 2009. The new implicit measures: an overview. In: Petty, R.E., Fazio, R.H., Brinol, P. (Eds.), *Attitudes: Insights from the New Implicit Measures*. Psychology Press, Hove, pp. 3–18.
- Pratto, F., Çidam, A., Stewart, A.L., et al., 2013. Social dominance in context and in individuals: contextual moderation of robust effects of social dominance orientation in 15 Languages and 20 Countries. *Soc. Psychol. Pers. Sci.* 4 (5), 587–599. <https://doi.org/10.1177/1948550612473663>.
- Pratto, F., Shih, H., 2000. Social dominance orientation and group context in implicit group prejudice. *Psychol. Sci.* 11 (6), 515–518. <https://doi.org/10.1111/1467-9280.00299>.
- Pratto, F., Sidanius, J., Levin, S., 2006. Social dominance theory and the dynamics of intergroup relations: taking stock and looking forward. *Eur. Rev. Soc. Psychol.* 17 (1), 271–320. <https://doi.org/10.1080/10463280601055772>.
- Pratto, F., Sidanius, J., Stallworth, L.M., Malle, B.F., 1994. Social dominance orientation: a personality variable predicting social and political attitudes. *J. Pers. Soc. Psychol.* 67 (4), 741–763. <https://doi.org/10.1037/0022-3514.67.4.741>.
- Rakić, T., Steffens, M.C., Mummendey, A., 2011. Blinded by the accent! The minor role of looks in ethnic categorization. *J. Pers. Soc. Psychol.* 100 (1), 16–29. <https://doi.org/10.1037/a0021522>.
- Rosseel, L., Speelman, D., Geeraerts, D., 2018. Measuring language attitudes using the personalized implicit association test: a case study on regional varieties of Dutch in Belgium. *J. Linguist. Geogr.* 6 (1), 20–39. <https://doi.org/10.1017/jlg.2018.3>.
- Rowatt, W.C., Franklin, L.M., Cotton, M., 2005. Patterns and personality correlates of implicit and explicit attitudes toward christians and muslims. *J. Sci. Stud. Relig.* 44 (1), 29–43. <https://doi.org/10.1111/j.1468-5906.2005.00263.x>.
- Sharma, D., Levon, E., Ye, Y., 2022. 50 years of British accent bias: stability and lifespan change in attitudes to accents. *Engl. World-Wide* 43 (2), 135–166. <https://doi.org/10.1075/eww.20010.sha>.
- Sibley, C.G., Wilson, M.S., Duckitt, J., 2007. Antecedents of men's hostile and benevolent sexism: the dual roles of social dominance orientation and right-wing authoritarianism. *Pers. Soc. Psychol. Bull.* 33 (2), 160–172. <https://doi.org/10.1177/0146167206294745>.
- Silverstein, M., 2003. Indexical order and the dialectics of sociolinguistic life. *Lang. Commun.* 23 (3/4), 193–229. [https://doi.org/10.1016/S0271-5309\(03\)00013-2](https://doi.org/10.1016/S0271-5309(03)00013-2).
- Stern, C., Axt, J.R., 2019. Group status modulates the associative strength between status quo supporting beliefs and anti-black attitudes. *Soc. Psychol. Pers. Sci.* 10 (7), 946–956. <https://doi.org/10.1177/1948550618799067>.
- Strongman, K.T., Woosley, J., 1967. Stereotyped reactions to regional accents. *Bri. J. Soc. Clin. Psychol.* 6 (3), 164–167. <https://doi.org/10.1111/j.2044-260.1967.tb00516.x>.
- Strycharczuk, P., Lopez-Ibanez, M., Brown, G., Leemann, A., 2020. General Northern English. exploring regional variation in the north of England with machine learning. *Front. Artif. Intell.* 3. <https://doi.org/10.3389/frai.2020.00048>.
- Sumantry, D., Choma, B.L., 2021. Accent-based stereotyping, prejudice, and their predictors. *Pers. Individ. Dif.* 179, <https://doi.org/10.1016/j.paid.2021.110894> 110894.
- Taylor, P.J., 1991. The English and their Englishness: 'a curiously mysterious, elusive and little understood people'. *Scott. Geogr. Mag.* 107 (3), 146–161. <https://doi.org/10.1080/00369229118736824>.
- Vari, J., Tamburelli, M., 2020. Standardisation: bolstering positive attitudes towards endangered language varieties? Evidence from implicit attitudes. *J. Multiling. Multicul. Dev.* 44 (6), 447–466. <https://doi.org/10.1080/01434632.2020.1829632>.
- Wales, K., 2000. North and South: an English linguistic divide. *Engl. Today* 16 (1), 4–15. <https://doi.org/10.1017/S026078400011378>.
- Wales, K., 2006. *Northern English: A Social and Cultural History*. Cambridge University Press, Cambridge.
- Wells, J.C., 1982. *Accents of English: Volumes 1-3, vol. 1*. Cambridge University Press, Cambridge.